

# Constitutive model of secondary annealing behavior of interconnected Cu in Cu/SiO<sub>2</sub> hybrid bonding

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**Abstract:** The three-dimensional (3D) packaging high-density interconnect technology, based on Cu/SiO<sub>2</sub> hybrid bonding, achieves Cu/Cu interconnects through Cu atomic diffusion during high-temperature annealing, enabling interconnect spacing reduction to below 10 μm. Variations in annealing conditions induce distinct microstructures and deformation behaviors in the interconnected Cu, which critically influence mechanical properties such as elastic modulus and yield stress. The mechanical performance of Cu plays an important role in the thermomechanical reliability of Cu/SiO<sub>2</sub> hybrid bonding.

To investigate the deformation behavior of interconnected Cu, this study employed nanoindentation experiments combined with finite element inverse analysis to establish power-law constitutive models for both the as-bonded Cu and the Cu subjected to secondary annealing. A softening effect was observed in the secondary annealed Cu. In addition, the microstructure of interconnected Cu was characterized by electron backscatter diffraction (EBSD), and it was found that, compared with the samples after initial bonding annealing, the grain size of Cu increased and the proportion of recrystallized grains increased after secondary annealing.

The constitutive model of interconnected Cu was derived, and it was found that both hardness and yield strength comply with the Hall-Petch relationship. The refinement of the constitutive model is of great significance for the deformation calculation of interconnected Cu in Cu/SiO<sub>2</sub> hybrid bonding structures and for improving the thermomechanical reliability of devices.

**Keywords:** secondary annealing

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,Cu/SiO<sub>2</sub> Hybrid bonding, Constitutive model, Nanoindentation, Microstructure

## 1. Introduction

With the rapid advancement of semiconductor technology, the functions of various integrated circuits are diversifying. Applications in home office, artificial intelligence (AI), and automotive electronics are driving the development of integrated circuits toward higher performance, higher integration, and higher reliability. As Moore's Law for semiconductors approaches its limits, front-end chip manufacturing has reached a technical bottleneck, making it difficult to increase the number of transistors per unit area while maintaining costs. As a result, the international community has begun to focus on advanced packaging technology, aiming to break through the limits of Moore's Law through high-density interconnect technology in the back-end process [1]. In emerging packaging technologies, Cu/SiO<sub>2</sub> hybrid bonding is a very promising technology. Cu/SiO<sub>2</sub> hybrid bonding does not require metal leads or microbumps, but achieves direct bonding through the interface between Cu/Cu and SiO<sub>2</sub>/SiO<sub>2</sub>. The bonding interconnections between the SiO<sub>2</sub> media provide mechanical support and electrical isolation for the entire integrated unit, while the vertical electrical interconnections of the chips are achieved through mutual bonding between the metal Cu bump. Due to the potential of Cu/SiO<sub>2</sub> hybrid bonding technology in achieving ultra-high density (interconnection pitch < 10 μm) electrical interconnection, it is considered an ideal bonding solution for high-bandwidth memory (HBM) [2]. Currently, there are several methods for achieving

Cu/SiO<sub>2</sub> hybrid bonding, including surface activation, direct bonding interconnect (DBI), and surface activation bonding (SAB).

The main process flow of Cu/SiO<sub>2</sub> hybrid bonding has been basically clarified<sup>[3,4]</sup>, (a):CMP polishing of the Cu/SiO<sub>2</sub> surface to form dishing on the Cu bump surface; (b):cleaning and modification of the surface to be bonded (removal of oxides, adsorption of specific functional groups, etc.) to make the surface to be bonded easy to bond and reduce the bonding temperature; (c):alignment of the upper and lower wafers;(d):Pre-bonding is performed at room temperature, and a weak connection is formed between the SiO<sub>2</sub> interfaces through van der Waals forces at room temperature; (e):Annealing, hydrogen bonds between the connected SiO<sub>2</sub> interfaces are dehydrated to form covalent bonds with greater energy, improving the bonding strength. At the same time, under the constraints of the surrounding SiO<sub>2</sub>, Cu bump expands and squeezes each other, and the atoms on the Cu/Cu contact surface diffuse sufficiently to form a reliable connection.

The ideal Cu/SiO<sub>2</sub> hybrid bonding structure can achieve a seamless bonding interface after annealing, thereby ensuring high thermal/mechanical reliability<sup>[5,6]</sup>. However, due to the alignment accuracy issues of existing processes, there are more complex interfaces in the entire Cu/SiO<sub>2</sub> hybrid bonding system, including Cu/Cu, Cu/SiO<sub>2</sub>, and SiO<sub>2</sub>/SiO<sub>2</sub> interfaces. Due to differences in the CTE of various materials and the presence of dishing on the Cu surface, deformation and residual stresses are introduced at the material interfaces during annealing, leading to the formation of microcracks and microvoids at the Cu/Cu and Cu/SiO<sub>2</sub> interfaces, significantly increasing the risk of potential failure. Juang<sup>[7,8]</sup> et al. observed that different annealing temperatures and times affect the growth of Cu grains, which in turn affects the quality of the bonding surface. Shie et al.<sup>[9]</sup> found that insufficient annealing time may adversely affect the bonding interface quality. Panckenko et al.<sup>[10,11]</sup> studied the changes in Cu grains after different reliability tests and found that Cu grains become significantly coarser after storage at high temperatures of 300-400°C. The

evolution of these microstructures is accompanied by the failure of the Cu/SiO<sub>2</sub> hybrid bonding at the macroscopic level. Shie et al.<sup>[7,8]</sup> found that, under a temperature cycle load of -55°C to 125°C, the Cu/Cu interface undergoes through-cracking failure.

The failure mechanism of the Cu/Cu interface under thermal fatigue loading is closely related to its mechanical parameters such as elastic modulus and yield stress. The elastic modulus determines the stiffness of the material in resisting thermal deformation: when the elastic modulus is high, the interface accumulates greater thermal stress (e.g., due to thermal expansion mismatch) during temperature cycling, accelerating the initiation of fatigue cracks; while yield stress influences the material's plastic response, and if the yield stress is low, thermal stresses are more likely to induce plastic deformation, which can mitigate stress concentration through energy dissipation; however, excessive plastic strain may also lead to damage accumulation. Conversely, if the yield stress is too high, the interface tends to release stresses in a brittle manner, promoting early cracking. Therefore, measuring the mechanical properties of the Cu/Cu interface has become a key focus in studying its thermal fatigue failure mechanisms, but due to the submicron structure of Cu/SiO<sub>2</sub> hybrid bonding, traditional mechanical testing methods such as tensile, compression and shear tests are no longer applicable. Nanoindentation is a simple and efficient method for evaluating the mechanical properties of materials. It can characterise the mechanical properties of sample surface materials at the micron and even nanometre scale and has high repeatability. The method proposed by Oliver and Pharr<sup>[14]</sup> is currently the most widely used nanoindentation test method. Okoro et al.<sup>[15]</sup> used the nanoindentation method to study the mechanical properties of TSV-Cu and analysed the influence of its microstructure on its performance. Ou et al.<sup>[16]</sup> used nanoindentation technology to measure the hardness of Cu in a Cu/SiO<sub>2</sub> hybrid bonding structure.

The constitutive relationship of a material is determined by the stress-strain curve obtained from experiments. For nanoindentation, it is impossible to directly characterise the elastic-plastic constitutive

relationship of a material. However, inspired by the research of Dao et al.<sup>[17]</sup>, finite element inversion combined with nanoindentation experiments can be used to derive the actual stress-strain curve of a material. The principle is that materials with the same elastic modulus, characteristic stress, and characteristic strain will produce the same load-displacement curves. By iteratively adjusting the material parameters in the finite element simulation, the load-displacement curves obtained from the simulation match those from the nanoindentation experiments, and the parameters obtained from the inversion are used to determine the material's true stress-strain curve. Long et al.<sup>[18]</sup> used this method to obtain the constitutive model of sintered silver particles; Zhang et al.<sup>[19]</sup> studied the constitutive behavior of TSV-Cu under different holding times using a combination of finite element simulation and nanoindentation experiments.

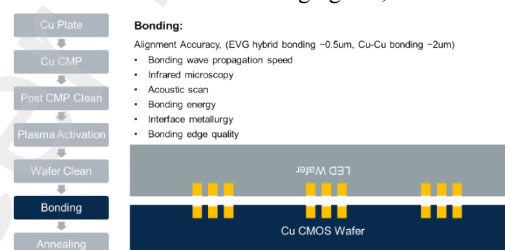
The paper focuses on the evolution of interconnected Cu mechanical properties in Cu/SiO<sub>2</sub> hybrid bonding structures before and after secondary annealing. Using nanoindentation experiments combined with finite element inversion, the elastic-plastic constitutive behavior of interconnected Cu under different secondary annealing conditions is derived, and its microstructural evolution is analysed. The aim is to provide theoretical guidance on the thermomechanical reliability of Cu/SiO<sub>2</sub> hybrid bonding structures improvement.

## 2. Experiments

### 2.1. The Preparation of Cu/SiO<sub>2</sub> hybrid bonding structures

The manufacturing process for bonding wafers is as follows: First, a 3.5 $\mu$ m silicon oxide layer is deposited on the clean wafer surface using PECVD, then chemically mechanically polished to level the surface and reduce the thickness of the silicon oxide layer to 3 $\mu$ m. Subsequently, photoresist is coated onto the silicon oxide surface, and the pad patterns are transferred using photolithography. Using plasma etching, recesses for the pads are etched into the silicon oxide surface. Subsequently, copper pads are filled using wafer electroplating. Finally, chemical

mechanical polishing is performed to remove excess copper and the seed layer from the wafer surface and control the surface morphology of the wafer. In the wafer bonding stage, the surface is first activated by plasma, and then the wafer surface is wetted to make it hydrophilic. After the upper and lower wafers are optically aligned, the middle of the wafers are first pre-contacted through the middle pin, and then, under the action of van der Waals force, the bonding wave diffuses, and the two wafers can be bonded at room temperature. Finally, annealing is performed at 300°C to complete the bonding, promoting the fusion of the SiO<sub>2</sub>/SiO<sub>2</sub> and Cu/Cu hybrid bonding interface. The process flow is shown in Fig. 1, and the sample structure is shown in Fig. 2. The above steps are completed by National Center for Advanced Packaging Co., Ltd.



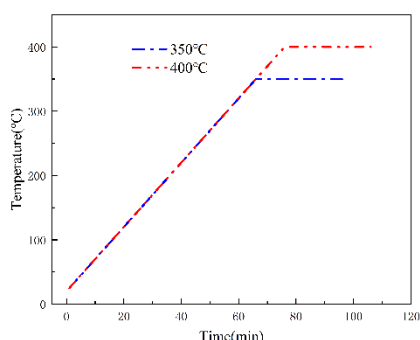
**Fig. 1.** process flow

The bonding chips cut from the wafer are 3 × 3 mm in size, with Cu pads 6  $\mu$ m in diameter and spaced 10  $\mu$ m apart, as shown in the Fig. 2. The cut chips are sealed with epoxy resin and then ground on a metallographic grinding and polishing table with #1200, #2000, and #3000 sandpaper in sequence until the Cu/Cu bonding cross section is exposed. Subsequently, polishing was performed using a mixture of Al<sub>2</sub>O<sub>3</sub> powder with a particle size of 0.5 $\mu$ m and deionised water, followed by a suspension of SiO<sub>2</sub> with a particle size of 0.05 $\mu$ m.

### 2.2. Secondary annealing experiment

In order to study the effect of secondary annealing on the mechanical properties of Cu/SiO<sub>2</sub> hybrid bonding, Cu/SiO<sub>2</sub> hybrid bonding samples were placed in a GSL-1750 single-zone tube electric furnace for annealing experiments. Considering that Cu/Cu bonding requires a high temperature of 400°C for sufficient atomic diffusion<sup>[20-22]</sup>, two different secondary annealing temperatures, 350°C and 400°C

were designed. The annealing temperature was raised from room temperature (25°C) to the annealing temperature, with a heating rate of 5°C/min. Argon was used as the protective gas. The samples was held at the annealing temperature for 30 minutes, then cooled naturally to room temperature. The process temperature curve is shown in Fig.3.



**Fig.3.** process temperature curve

### 2.3. Mechanical properties testing

In order to eliminate the influence of surface roughness on the experimental test and ensure better repeatability of the experiment, the Leica EM TIC 3X ion beam milling system was used to further process the polished Cu/Cu interface cross-section. After processing, the surface roughness was kept less than 10 nm, and then the mechanical properties of the interconnected Cu cross-section of the Cu/SiO<sub>2</sub> hybrid bonding sample were tested using a G200 nanoindenter manufactured by Agilent Nano Indenter. The indenter was a Berkovich indenter, and the load/displacement curves, as well as the variation curves of elastic modulus and hardness with depth, were obtained using the continuous stiffness method (CSM)<sup>[14]</sup>. The mechanical properties of interconnected Cu were investigated via nanoindentation before and after

secondary annealing, with the maximum indentation depth set to 300 and the strain rate at 0.05s<sup>-1</sup>. In order to ensure the accuracy of the data, six test points were selected for indentation testing in each group of experiments. The experimental results of the samples after initial bonding are shown in Fig.4. It can be seen that the elastic modulus and hardness values tend to stabilise at 200-300 nm, which confirms the repeatability of the experiment. The average elastic modulus and hardness of each test point in this range were calculated as parameters for subsequent simulation.

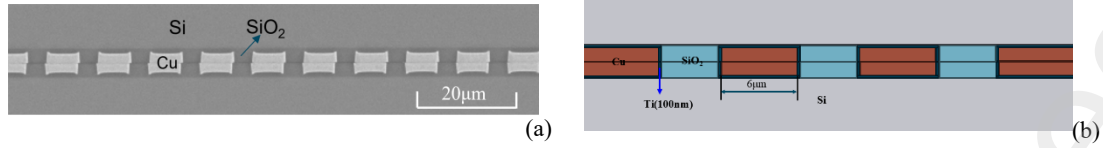
### 2.4. Microstructural characterization

Due to the stringent requirements of EBSD experiments on the sample surface to be tested, any contamination or deformation layer on the surface of the sample will affect the accuracy of the EBSD diffraction patterns<sup>[23]</sup>. Therefore, the Leica EM TIC 3X ion beam milling system was used to perform ion milling on the polished cross-sections of the samples under conditions of 6 kV for 20 min, 5 kV for 20 min, and 4 kV for 20 min, in order to remove deformation layers and chemical reagents introduced during the grinding and polishing process.

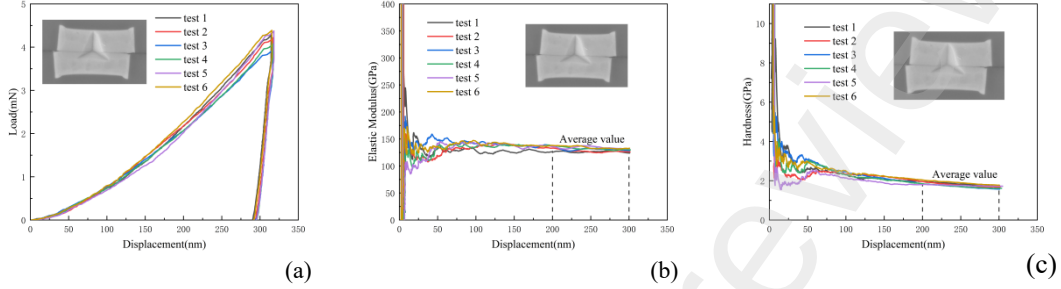
EBSD observations were conducted on the prepared cross-sections using a Zeiss Merlin high-performance field emission scanning electron microscope. The environmental scanning voltage was set to 20 kV, and the acceleration current was 5 nA. The lattice parameters of the phase to be scanned are provided in Table 1. To enhance the resolution of the EBSD data acquisition patterns, the cross-section to be measured was positioned at a 70° angle relative to the incident electron beam. Subsequently, Channel 5 software was used to analyse the microstructure of the interconnected Cu based on the EBSD images obtained.

**Table 1** lattice parameters

| Crystal Phase | a     | b     | b     | Alpha  | Beta   | Gamma  | space group | database |
|---------------|-------|-------|-------|--------|--------|--------|-------------|----------|
| Copper        | 3.61Å | 3.61Å | 3.61Å | 90.00° | 90.00° | 90.00° | 225         | HKL      |



**Fig.2.** (a) SEM image of the cross section of the Cu/SiO<sub>2</sub> hybrid bonding structure after initial bonding, (b) The schematic diagram of Cu/SiO<sub>2</sub> hybrid bonding



**Fig.4.** (a) Load-displacement curve, (b) Modulus-displacement curve, (c) Hardness-displacement curve

### 3. Derivation algorithm for power-law constitutive relations of interconnected Cu

For metallic materials, the power-law type elastic-plastic stress-strain constitutive model is generally represented by the following Eq.(1):

$$\sigma = \begin{cases} E\varepsilon & \sigma \leq \sigma_y \\ \sigma_y \left( 1 + \frac{E}{\sigma_y} \varepsilon_p \right)^n & \sigma > \sigma_y \end{cases} \quad (1)$$

Where,  $E$  is the elastic modulus of the plastic material,  $\sigma_y$  is the yield strength,  $\varepsilon_p$  is the plastic strain and  $n$  is the hardening exponent. It can be observed that a linear relationship can be established among stress, strain, and elastic modulus when the applied stress is below the yield strength. When the stress is greater than the yield strength, the relationship can be defined as a power law relationship.

The stress-strain relationship of metallic materials is shown by the red curve in Fig.5(a), which intersects with the blue curve representing the ideal elastic-plastic constitutive model at a special point  $(\varepsilon_r, \sigma_r)$ , where  $\varepsilon_r$  is the characteristic strain and  $\sigma_r$  is the characteristic stress. For materials with the same elastic modulus, as long as their stress-strain curves pass through this point, consistent load-displacement curves can be obtained in nanoindentation tests<sup>[24]</sup>. As shown in Fig.5(b), the black curve represents the typical load/displacement curve of a nanoindentation test on an elastoplastic

material, which is typically divided into three stages: the loading stage, the holding stage, and the unloading stage. In the figure,  $P_{\max}$  denotes the maximum load encountered during the experiment, which occurs at the maximum penetration depth  $h_m$ . After the load is completely unloaded, the plastic deformation of the material results in the formation of residual indentation depth  $h_r$ . Simultaneously, the load head performs work during the loading process, and the total work done is denoted  $W_t$ , and  $W_t = W_e + W_p$ ; where  $W_e$  represents the elastic work that can be recovered during unloading, expressed as the area enclosed by the loading curve, unloading curve, and the horizontal axis.  $W_p$  is the plastic work stored in the material that cannot be recovered, represented by the area enclosed by the unloading curve, the horizontal axis, and the red dashed line in the figure.

In order to obtain the stress/strain relationship of interconnect Cu, this study used commercial software ABAQUS to perform finite element inversion analysis, thereby determining the constitutive model of interconnect Cu. The finite element model adopted is shown in Fig.6. A two-dimensional axisymmetric model was established. Since the width of the blocking layer Ti (100nm) is much smaller than the diameter of the interconnect Cu (6μm), its influence on the inversion results can be neglected. The Cu and SiO<sub>2</sub> interfaces are connected via common nodes. To ensure the accuracy of the inversion results, the displacement

and boundary conditions must align with the experimental conditions. Therefore, symmetric constraints are applied to the left and bottom sides of the model, and vertical constraints are applied to the bottom of the model. The element type is CAX4R, with a total of 10,464 elements. Since the indenter used in this nanoindentation experiment is a standard Berkovich indenter, according to the same projection rule, it can be simplified to a conical indenter with a half-cone angle of  $70.3^\circ$  in finite element model [25] to reduce modelling complexity. The indenter is treated as a rigid material, and contact must be defined between it and the interconnected Cu surface, with a friction coefficient of 0.1. The elastic modulus and Poisson's ratio of  $\text{SiO}_2$  are taken as 73 GPa and 0.17, respectively [26].

The power-law constitutive model of interconnected Cu is shown by Eq (1), where  $E$  can be directly obtained from nanoindentation testing. By obtaining the values of  $\sigma_y, \epsilon_p$ , and  $n$  and solving the equation, the constitutive model for interconnected Cu can be determined. Fig.7 shows the flowchart for inverting the constitutive model of interconnected Cu. The specific inversion process is as follows.

Materials with the same elastic modulus and characteristic stress can obtain the same loading curve during the loading stage [24], thus  $\sigma_r$  can be first calculated using the Eq(2) proposed by Antunes et al. [27].

$$\frac{E_r}{H} = 0.231 \left( \frac{E_r}{\sigma_r} \right) + 4.910 \quad (2)$$

Where  $H$  is the hardness of the interconnected Cu obtained through nanoindentation testing,  $E_r$  is the shear modulus and can be obtained using Eq (3).

$$\frac{1}{E_r} = \frac{1-\nu^2}{E} + \frac{1-\nu_i^2}{E_i} \quad (3)$$

Where  $E$  and  $\nu$  represent the elastic modulus and Poisson's ratio of the interconnected Cu, respectively.  $E$  obtained from experiments and the value of  $\nu$  is 0.3.  $E_i$  and  $\nu_i$  are the elastic modulus and Poisson's ratio of the diamond indenter, respectively, are taken as 1141 GPa and 0.07 [28].

After calculating the initial characteristic stress  $\sigma_r$ , the influence of the hardening exponent  $n$  can be temporarily ignored by setting  $n = 0$ , and the ideal elastic-plastic constitutive model shown by the blue curve in Fig.5(a) can be used for inversion. This yields load/displacement curves similar to those obtained from nanoindentation experiments, as well as the maximum inverted load  $P_{\max}^{FEM}$ . Since both are influenced by the same factors, therefore, by continuously adjusting the value of  $\sigma_r$ , the alignment of the load/displacement curve obtained from inversion with the curve obtained from actual nanoindentation testing can be observed, thereby determining the final characteristic stress  $\sigma_r$ . During the adjustment process, the maximum simulated load  $P_{\max}^{FEM}$  obtained from finite element inversion is continuously compared with the maximum experimental load  $P_{\max}^{EXP}$  obtained from actual nanoindentation testing. This process is repeated until the relative error between the two is less than 0.5%, at which point the final characteristic stress  $\sigma_r$  can be determined. During the iterative calculation process,  $\sigma_r$  is reassigned using Eq (4) [29].

$$\sigma_r(i+1) = \sigma_r(i) \frac{P_{\max}^{EXP}}{P_{\max}^{FEM}} \quad (4)$$

Where  $\sigma_r(i)$  and  $\sigma_r(i+1)$  represent the characteristic stresses at the  $i$ -th and  $i+1$ -th iterations, respectively.

After determining  $\sigma_r$ , the hardening exponent  $n$  can be solved using the dimensionless function expressions (5) and (6) given by Dao et al. [17].

$$\frac{W_p}{W_t} = 1.61217 \left\{ 1.13111 - 1.74756 \left[ -1.49291 \left( \frac{h_r}{h_m} \right)^{2.535334} \right] - 0.075187 \left( \frac{h_r}{h_m} \right)^{1.135826} \right\} \quad (5)$$

Where  $W_p$  is the plastic work done by the indenter during the nanoindentation test, and  $W_t$  is the total work done by the indenter.

$$\frac{h_r}{h_m} = A \left[ \ln \left( \frac{\sigma_r}{E_r} \right) \right]^3 + B \left[ \ln \left( \frac{\sigma_r}{E_r} \right) \right]^2 + C \left[ \ln \left( \frac{\sigma_r}{E_r} \right) \right] + D \quad (6)$$

The values of A, B, C, and D are as follows:

$$A = 0.010100n^2 + 0.0017639n - 0.0040837$$

$$B = 0.14386n^2 + 0.018153n - 0.088198$$

$$C = 0.59505n^2 + 0.034074n - 0.65417$$

$$D = 0.58180n^2 - 0.088460n - 0.67290$$

However, calculations revealed that even a small change in  $\frac{\sigma_r}{E_r}$  would cause a significant change in the hardening exponent  $n$ . Therefore,  $n$  calculated at this point is only used to determine the characteristic strain  $\varepsilon_r$ . Once  $\varepsilon_r$  is determined, the value of  $n$  must be further adjusted to determine the final constitutive model.  $\sigma_r$  and  $\varepsilon_r$  are not affected by  $n$ <sup>[30]</sup>.

The initial estimate can be obtained by using the Eq (7) provided by Lee et al.<sup>[31]</sup>, as shown below

$$\varepsilon_r = \exp\left(\frac{166.7}{E_r / \sigma_r + 177.3} - 3.91\right) \quad (7)$$

The elastoplastic constitutive equation, defined by the initial characteristic strain  $\varepsilon_r$ , the derived characteristic stress  $\sigma_r$ , and hardening exponent  $n$ , was implemented in ABAQUS for finite element analysis. The value of  $\varepsilon_r$  also affects the load/displacement curve and the maximum inverted load  $P_{\max}^{FEM}$  obtained through inversion, so the solution method is consistent with that of  $\sigma_r$ . By iteratively assigning values to  $\varepsilon_r$ , until the relative error between the maximum simulated load  $P_{\max}^{FEM}$  obtained from inversion and the maximum experimental load  $P_{\max}^{EXP}$  obtained from the

actual nanoindentation test is less than 0.5%. At this point, the final characteristic strain  $\varepsilon_r$  can be determined. The iterative formula for  $\varepsilon_r$  is shown in Eq(8).

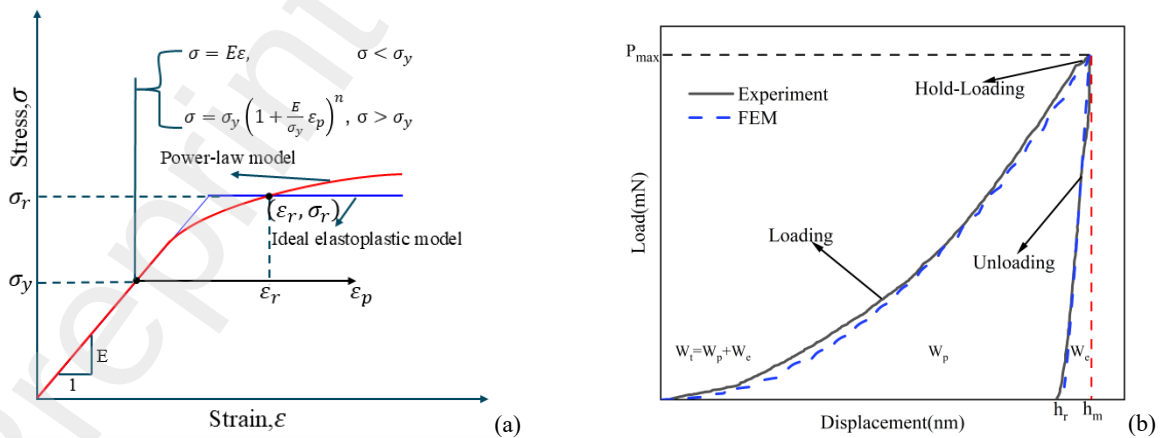
$$\varepsilon_r(i+1) = \varepsilon_r(i) \frac{F_{\max}^{FEM}}{F_{\max}^{EXP}} \quad (8)$$

As illustrated in Fig. 6(b) and Fig. 6(c), the Von Mises stress distributions of the interconnected Cu are depicted at the maximum loading depth and after unloading, respectively. The maximum values appear directly below the indenter.

According to Antunes et al.<sup>[27]</sup>, the unloading portion of the load/displacement curve is controlled by the control, so the overlap between the two curves should be improved by continuously adjusting the value until the two curves are almost coincident, as shown in Fig.5(b). This means that the material parameters used in the finite element simulation are consistent with the experiment. Substituting the final values of  $\sigma_r, \varepsilon_r$  and  $n$  into the nonlinear expression in Eq (1) yields Eq(9).

$$\sigma_r = \sigma_y \left(1 + \frac{E}{\sigma_y} \varepsilon_r\right)^n \quad (9)$$

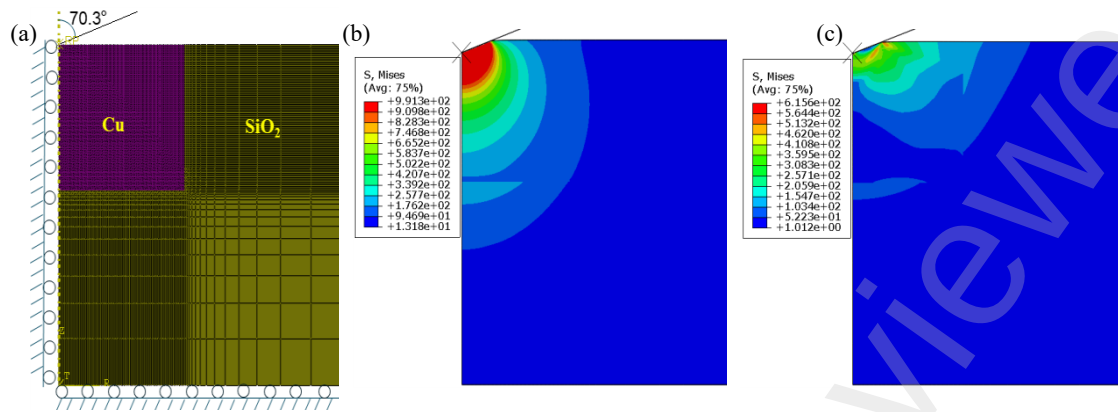
At this point, the yield strength of the interconnected Cu can be confirmed, and the power-law constitutive equation for the interconnected Cu can finally be obtained.



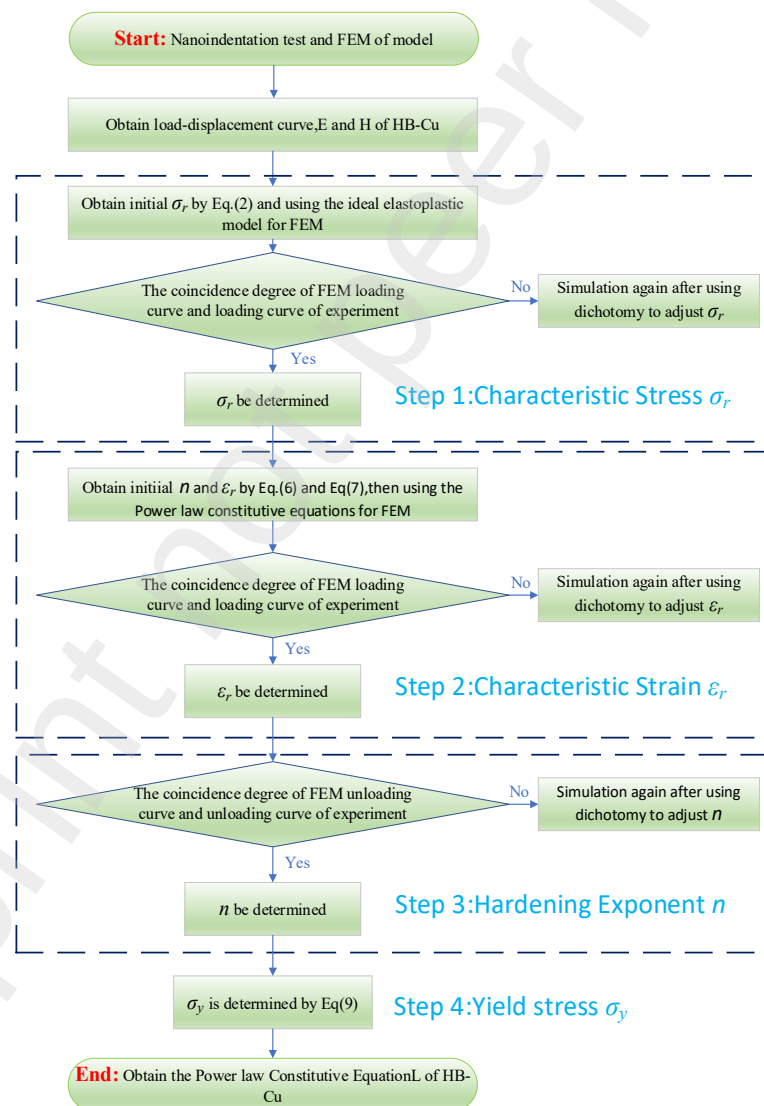
**Fig.5.** (a) Elastic-plastic constitutive model of metal materials, (b) Load-displacement curves of nanoindentation



experiments and finite element simulation curves



**Fig.6.** (a)Finite element model after initial bonding, (b) Von Mises stress contour map at maximum load depth, (c) Von Mises stress cloud map after unloading



**Fig.7.** Flowchart for determining the power-law type constitutive relationship of interconnected Cu



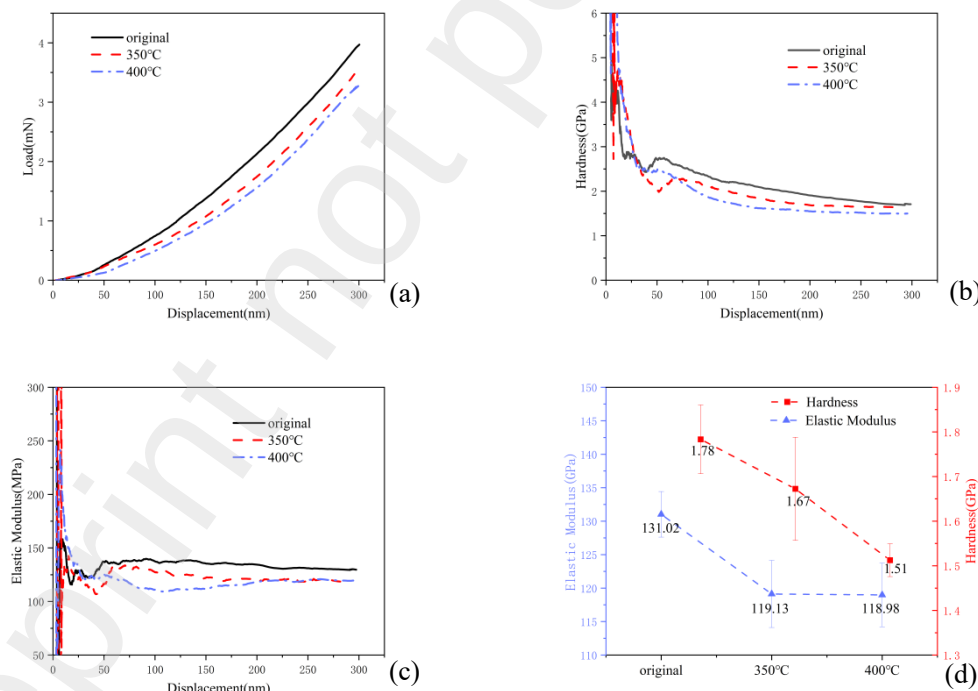
## 4. Results and discussion

### 4.1. Constitutive behavior of interconnected Cu before and after secondary annealing

Fig.8 shows the nanoindentation response curves of interconnected Cu in the initial bonding state and at secondary annealing temperatures of 350°C and 400°C. The displacement-load curves of interconnected Cu after different secondary annealing in Fig.8(a) show that, at the same indentation depth, the load required for secondary annealing is significantly lower than that for the bonding state samples that did not undergo secondary annealing. Additionally, as shown in Fig.8(b) and Fig.8(c), which present the changes in hardness and elastic modulus with displacement, respectively, when the indentation depth reaches 200–300nm, the mechanical parameter responses stabilise. The stable hardness and elastic modulus results are shown in Fig.8(d), where it can be observed that compared with

the bonding state samples, secondary annealing at 350°C and 400°C caused the elastic modulus to decrease from 131.2GPa to 119.13GPa and 118.98GPa, and the hardness to decrease from 1.78GPa to 1.67GPa and 1.51GPa, which means that the interconnected Cu exhibited softening behavior after secondary annealing. and this softening behavior becomes more pronounced as the secondary annealing temperature increases from 350°C to 400°C.

The power-law constitutive relationships under different states are shown in Table 2 and Fig.9. It can be seen that compared with the bonding state samples, the yield strength of interconnected Cu decreased from 70.52MPa to 60.54MPa and 56.12MPa after secondary annealing at 350°C and 400°C, respectively. This means that secondary annealing can lead to a decrease in the yield strength of interconnected Cu, and the decrease becomes more significant as the secondary annealing temperature increases.



**Fig.8.** Nanoindentation curves under different conditions (a) Load-displacement curve, (b) Hardness-displacement curve, (c) Modulus-displacement curve, (d) Hardness and modulus distribution

**Table 2** Power-law constitutive parameters of interconnected Cu in different states

| State    | Elastic modulus,E (GPa) | Hardness, H (GPa) | Representative stress, $\sigma_r$ (MPa) | Representative strain, $\varepsilon_r$ | Yield strength $\sigma_y$ (MPa) | Hardening exponent,n |
|----------|-------------------------|-------------------|-----------------------------------------|----------------------------------------|---------------------------------|----------------------|
| Original | 131.02 $\pm$ 3.41       | 1.78 $\pm$ 0.08   | 467.8 $\pm$ 28.77                       | 0.0232 $\pm$ 0.001                     | 70.52 $\pm$ 9.77                | 0.4975 $\pm$ 0.002   |
| 350°C    | 119.13 $\pm$ 5.02       | 1.67 $\pm$ 0.11   | 399.5 $\pm$ 17.51                       | 0.023 $\pm$ 0.0006                     | 60.54 $\pm$ 2.66                | 0.5096 $\pm$ 0.001   |
| 400°C    | 118.98 $\pm$ 4.75       | 1.51 $\pm$ 0.03   | 417.68 $\pm$ 28.18                      | 0.024 $\pm$ 0.0008                     | 56.12 $\pm$ 4.36                | 0.4961 $\pm$ 0.01    |

#### 4.2. Microstructural changes in the morphology of the Cu interface before and after secondary annealing

Fig.10 shows the cross-sectional morphology of the interconnected Cu/Cu interface in the initial bonding state and after secondary annealing at 350°C and 400°C. It can be seen that there are no obvious defects in the Cu/Cu interface in the initial bonding state, but there are small voids distributed. These small voids (<100 nm) are caused by a variety of process factors. On the one hand, during the preparation of bonding structures, certain defect particles, such as silicon dust, polymer residues, or surface contamination caused by touch, are inevitably introduced onto the Cu pads. These can lead to the formation of voids at the Cu interconnection interface. On the other hand, in a study by Gondcharton et al.<sup>[32]</sup>, it was found that the roughness of the Cu pad surface after CMP also affects the formation of interface voids. In addition, A gap was formed at the bonding interface between the Cu pad and SiO<sub>2</sub>. This may be due to the significantly higher thermal expansion coefficient of Cu than SiO<sub>2</sub>. During annealing, as the temperature increased, the thermal mismatch between Cu and SiO<sub>2</sub> caused compressive stress on the Cu pad and tensile stress on SiO<sub>2</sub>. In addition, if the depression value of Cu before bonding is not appropriate or the alignment accuracy during bonding is insufficient<sup>[10]</sup>, defects may occur at the SiO<sub>2</sub> bonding interface near the Cu pad. Similar interface morphologies have been reported in studies by Ji et al.<sup>[33]</sup> and Fujino et al.<sup>[34]</sup>. After undergoing secondary annealing at 350°C and 400°C,

there was no obvious change in the morphology of the Cu/Cu bonding interface.

The microstructure of interconnected Cu structures before and after secondary annealing was studied by EBSD. Three interconnected Cu structures were characterised in the initial bonding state and in samples annealed at 350°C and 400°C. Fig.11 shows the EBSD results of the interconnected Cu in three different states. For the bonded samples, the average grain size of the interconnected Cu was 0.62  $\mu$ m, the average grain size after secondary annealing at 350°C was 0.67  $\mu$ m, and the average grain size after secondary annealing at 400°C was 0.71  $\mu$ m. Secondary annealing causes grain growth, and that increasing the secondary annealing temperature further enhances grain growth.

Similar to polycrystalline metals that undergo cold deformation, during annealing treatment, the microstructure of metals undergoes a evolution process<sup>[35]</sup>. The recrystallisation temperature of Cu is 200–250°C<sup>[36]</sup>, so during the annealing process, the microstructure of the Cu interconnects must undergo these three stages of evolution, as shown in Fig.12(a):

(1) Response stage: As the temperature gradually increases from room temperature (25°C), the Cu interconnect microstructure remains unchanged. However, thermal mismatch causes deformation of the Cu interconnect structure, leading to the accumulation of a large amount of strain within it. These strains are stored in the interconnect structure as deformation energy, which subsequently serves as the driving force for recrystallization.

(2) Recrystallisation stage: As the temperature continues to rise and reaches the recrystallisation

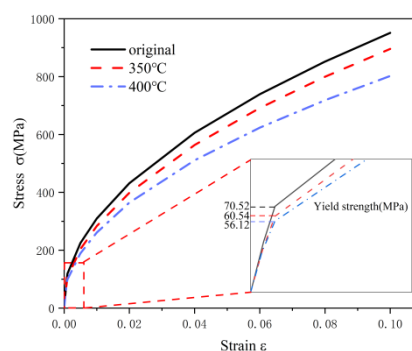
temperature range of Cu (200°C–250°C), nucleation begins at the triple points of grain boundaries. The release of strain energy drives the growth of nuclei, which macroscopically manifests as a grain refinement process. The literature indicates<sup>[36]</sup> that this stage can last for 60 min until recrystallisation reaches a stable state, at which point the grain size no longer changes.

(3) Grain growth stage: When the temperature exceeds 250°C, the recrystallisation stage ends. At this point, the temperature increase enhances grain boundary mobility, becoming the primary driving force for subsequent grain coalescence and growth. Macroscopically, this manifests as small grains merging together, resulting in grain growth.

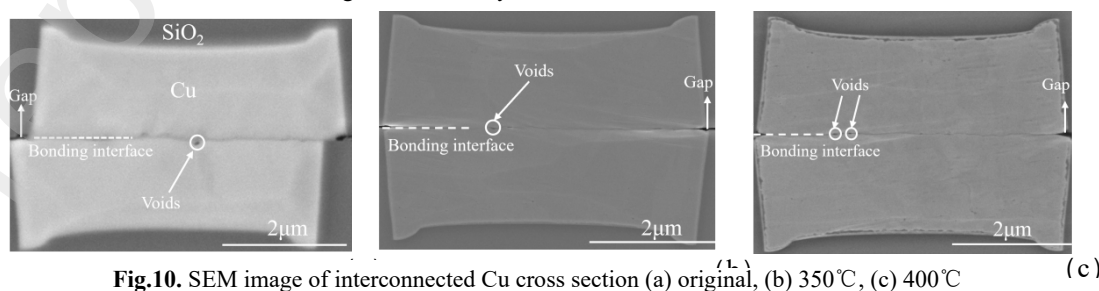
Fig.12(b) shows the local misorientation(KAM) difference of the interconnected Cu structure in the initial bonding state, after secondary annealing at 350°C and 400°C. It can be seen that the KAM value of the interconnected Cu structure decreased from 0.423 to 0.417 and 0.372 after secondary annealing, indicating that the deformation and dislocation inside the grains of the interconnected Cu structure were partially released after secondary annealing, so that the stored energy inside the grains can be fully released. The decrease is more significant after secondary annealing at 400°C than after secondary annealing at 350°C. At the same time, Fig.12(c) shows the grain boundary distribution of the interconnected Cu structure in the three states. It can be seen that after secondary annealing, the HAGB content in the interconnected Cu structure increased from 0.849 to 0.878 and 0.883, and the LAGB content decreased from 0.151 to 0.122 and 0.117. In addition, Fig.12(d) shows that after secondary annealing, the proportion of recrystallised grains in the interconnected Cu structure increased, and the proportion of subgrains decreased. This indicates that during the secondary

annealing process, the subgrains formed in the recovery stage gradually evolved into recrystallised grains surrounded by HAGB through the absorption of dislocations and grain boundary migration. This explains the grain coarsening observed after the secondary annealing and also shows that the recrystallisation process of the interconnected Cu structure is more complete after the secondary annealing and that increasing the secondary annealing temperature can accelerate the recrystallisation process.

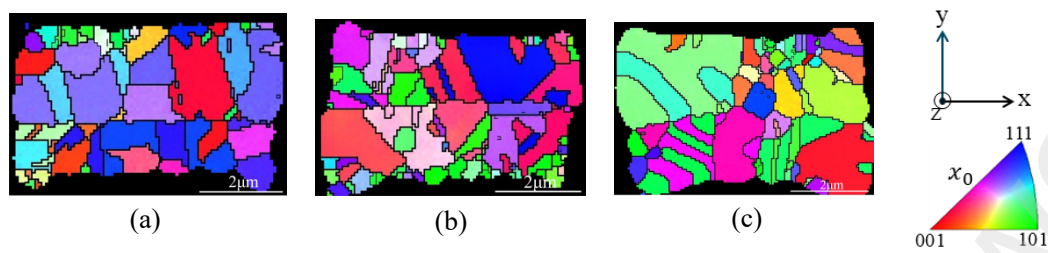
The grain boundary distribution in Fig.12(c) indicates that the bonding interface in the initial bonding state was relatively flat, indicating that the intergrowth of the two Cu pads at the interface was low. After secondary annealing, the non-planarity of the interconnected Cu interface was significantly improved, indicating that the grain intergrowth at the interface increased after secondary annealing, which was more significant at 400°C than at 350°C. The observable grain intergrowth mainly occurred at the triple intersection points. Panckenko et al.<sup>[11]</sup> also observed a similar phenomenon after storage at 400°C for 6 hours.



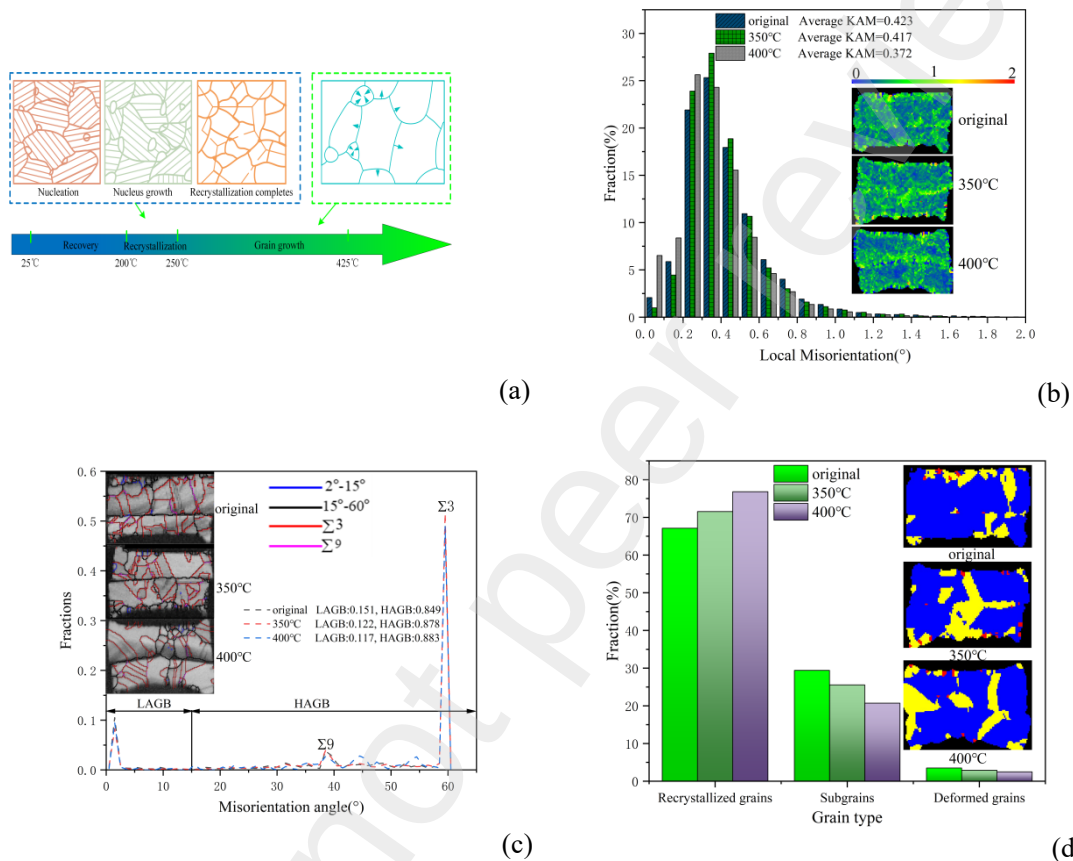
**Fig.9.** Power-law constitutive behavior of interconnected Cu in different states



**Fig.10.** SEM image of interconnected Cu cross section (a) original, (b) 350°C, (c) 400°C



**Fig.11.** EBSD images of interconnected Cu in different states (a) Original, (b) 350°C, (c) 400°C



**Fig.12.** (a) Changes in microstructure with increasing annealing temperature, (b) Distribution of local orientation differences of interconnected Cu in different states, (c) Distribution and proportion of grain boundary types in interconnected Cu in different states, (d) Recrystallisation distribution of interconnected Cu in different states

#### 4.3. Micro-macro mechanical responses of interconnected Cu before and after secondary annealing

Annealing triggers the evolution of the microstructure and significantly alters the macromechanical properties of Cu. Fig.13(a) shows the distribution of elastic modulus, hardness, and yield strength for three groups of samples with different grain sizes. It can be seen that as the grain size increases, the yield strength, hardness, and elastic modulus

decreases for all the three group of samples. However, the grain size dependence of yield strength and hardness is significantly stronger than that of the elastic modulus; the former two parameters obey the Hall-Petch relationship. Hardness and yield strength are inversely proportional to the square root of the average grain size  $D$ , as shown by Eq(10) and (11). This suggests that yield strength and hardness demonstrate a stronger dependence on grain size, whereas the elastic modulus is predominantly governed by residual stress. This difference can be explained by the competitive

effects of grain size and residual stress. Budiman<sup>[37]</sup> et al. found that residual stress within the material decreases after annealing, and the decrease in the KAM value in Fig. 12(b) further illustrates the reduction in residual stress in the interconnected Cu structure after secondary annealing. The reduction in residual stress is considered to be one of the reasons for the decrease in elastic modulus. In addition, it is also affected by grain orientation<sup>[38,39]</sup>, and a similar trend was found in the study by Li et al.<sup>[30]</sup>.

$$H = H_0 + \frac{K_1}{\sqrt{d}} \quad (10)$$

$$\sigma_y = \sigma_{y0} + \frac{K_2}{\sqrt{d}} \quad (11)$$

In this model,  $H_0$  and  $\sigma_{y0}$  are governed by initial dislocation density, while  $k_1$  and  $k_2$  represent material-specific coefficients. For the interconnected Cu in Hybrid Bonding, the fitted parameters  $H_0 = -2.34$ ,  $k_1 = 3.26$ ,  $\sigma_{y0} = -160.86$ , and  $k_2 = 182$  were obtained (Fig. 13b). After 300°C annealing, Hybrid Bonding Interconnected Cu exhibited a yield strength of 70.52 MPa, 68% higher than that of TSV-Cu (42.83 MPa [30]), indicating significantly enhanced resistance to plastic deformation.

## 5. Conclusions

In this study, nanoindentation experiments

combined with finite element inversion were employed to determine the power-law constitutive model of hybrid bonding interconnected Cu before and after secondary annealing. Simultaneously, the evolution of its microstructure was investigated, revealing the influence of microstructural characteristics on the constitutive behavior. The key conclusions are summarized as follows:

1. A downward trend in elastic modulus, hardness, and yield strength was observed in hybrid bonding interconnected Cu following secondary annealing, which was more pronounced after secondary annealing at 400°C.
2. After secondary annealing, the proportion of recrystallised grains in the interconnected Cu increased, and an increase in grain size was observed in all cases. This indicates that secondary annealing promotes the recrystallisation process. Additionally, after secondary annealing, the number of grain intergrowth at the interface increased significantly.
3. The hardness and yield strength decrease with increasing grain size, which is consistent with the Hall-Petch relationship. The constitutive relationship quantitatively reflects the annealing effect. The interconnected Cu exhibits higher yield strength than TSV-Cu after initial annealing at 300°C.

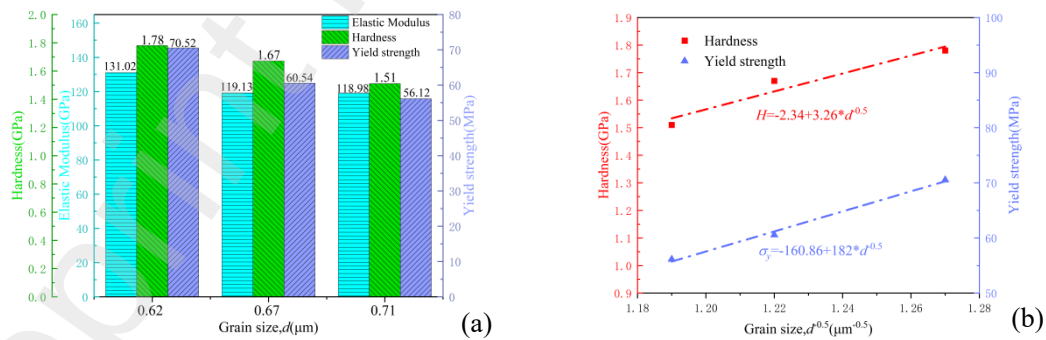


图 13 (a)不同晶粒尺寸的模量、硬度和屈服强度分布,(b)硬度、屈服强度的 Hall-Petch 方程拟合

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